An applied pull is not always the net force

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Driving force / ES: fuerza impulsora / PT: força motriz

The pull before resistance is subtracted.

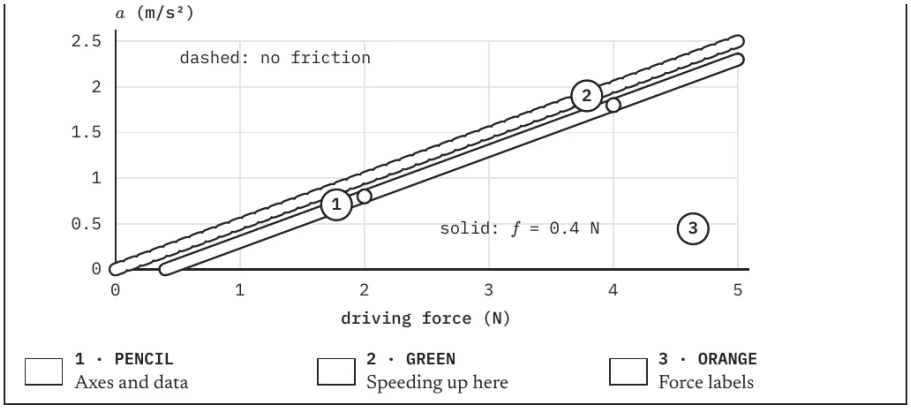
Also on your note page: Net force, Slope, Friction offset.

B READ THE RELATIONSHIP

Fnet = Fdrive − f; a = Fnet/M

As you read, find: driving pull · friction · net force.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg system slides with 0.4 N of opposing friction. Compare acceleration with driving forces of 2 N and 4 N .

Given M = 2 kg f = 0.4 N F drive = 2 N , then 4 N

Rule Subtract friction. Then a = F net / M .

Work a 1 = (2 − 0.4)/2 = 0.8 a 2 = (4 − 0.4)/2 = 1.8

Answer 0.8; 1.8 m/s² Driving force doubles; a does not.

E SAVE FOR THE FINAL MEETING

A proposed push on XJ must be added to every other external force. Only the net force belongs in the calculation of acceleration.

Read the physics: Driving force

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Keep the total moving mass the same. With a cart and hanger, move mass from the cart to the hanger. This changes the driving pull while keeping the moving mass constant.

02 HOW IT WORKS

The driving force is one pull. Subtract any opposing friction to find the net force. Divide net force by total mass to find acceleration.

03 WHAT TO CHECK

At fixed mass, doubling the net force doubles acceleration. Doubling the driving pull need not double acceleration when friction is present.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg system slides with 0.4 N of opposing friction. Compare acceleration with driving forces of 2 N and 4 N . 0.8; 1.8 m/s² Driving force doubles; a does not.

Model check: Fnet = Fdrive − f; a = Fnet/M.

05 CONNECT TO XJ

A proposed push on XJ must be added to every other external force. Only the net force belongs in the calculation of acceleration.

SAY IT BEFORE YOU WRITE

Which force goes in Newton's second law? The driving pull is ___. After friction, the net force is ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Driving force / ES: fuerza impulsora / PT: força motriz

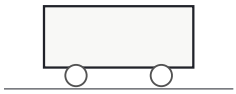
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, driving force means …	A rope pulls a crate while friction acts against it. Which arrow shows the driving force? The driving force comes from ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch the pull and opposing friction. Label driving force on one arrow and net force on a separate arrow.

Labels you can use: driving pull · friction · net force.



Tell a partner: The driving pull is ___. After friction, the net force is ___.

On assignment: Driving force

SECTION E / BACK / FINAL MEETING ONLY Name _____

A rehearsal uses the same asteroid model twice. The second net push is twice the first, for the same time. Operations needs a comparison.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Flight-test engineer**

Draw two net-force arrows and two velocity-change arrows. Recommend the stronger change and state what you kept fixed.

☐ **Test-range broadcaster**

Tell viewers which run changes velocity more. Ask your producer whether doubling a motor's pull always doubles net force.

Word bank: driving pull · friction · net force.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."

Editor's check: Distinguish a doubled applied force from a doubled net force.